

INF1004 Mathematics II

Tutorial 4: Eigenvalues & Eigenvectors

SIT

Question 1: Eigenvalues and Eigenvectors

Find the eigenvalues and eigenvectors for the following matrices:

$$A = \begin{pmatrix} 4 & 2 \\ 3 & 3 \end{pmatrix} \quad B = \begin{pmatrix} 7 & -3 \\ -2 & 2 \end{pmatrix}$$

$$C = \begin{pmatrix} 3 & 1 \\ -2 & 0 \end{pmatrix} \quad D = \begin{pmatrix} -2 & -2 \\ -5 & 1 \end{pmatrix}$$

Question 2: Diagonalization (2x2)

Diagonalize the following matrices. That is, find an invertible matrix P and a diagonal matrix D such that each matrix is expressed as PDP^{-1} .

$$A = \begin{pmatrix} 4 & 5 \\ -1 & -2 \end{pmatrix}$$

$$B = \begin{pmatrix} 7 & -2 \\ -1 & 8 \end{pmatrix}$$

Question 3: Diagonalization (3x3)

Diagonalize the following matrix:

$$A = \begin{pmatrix} 1 & 0 & 4 \\ 0 & 2 & 0 \\ 3 & 1 & -3 \end{pmatrix}$$

Question 4: Eigenvector Verification

Show that $\vec{v}$ is an eigenvector of matrix A :

$$A = \begin{pmatrix} -1 & 1 & 2 \\ -6 & 2 & 6 \\ 0 & 1 & 1 \end{pmatrix}, \quad \vec{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Find the other eigenvectors and hence diagonalize A .

Question 5: Eigenvector Verification

Show that $\vec{v}$ is an eigenvector of matrix A :

$$A = \begin{pmatrix} -5 & 8 & 32 \\ 2 & 1 & -8 \\ -2 & 2 & 11 \end{pmatrix}, \quad \vec{v} = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$$

Find the other eigenvectors and hence diagonalize A .

6. The "Mini-Internet" Calculation

The Scenario: Imagine a tiny internet with only 2 websites: **Site A** and **Site B**.

- Users on Site A have a 50% chance of clicking to Site B (and 50% stay).
- Users on Site B have a 100% chance of clicking to Site A (0% stay).

This creates the "Link Matrix" M :

$$M = \begin{pmatrix} 0.5 & 1 \\ 0.5 & 0 \end{pmatrix}$$

Note: Columns sum to 1, representing probability).

The Task: Find the **Steady State Vector** $\vec{r}$.

- One of the eigenvalues is 1, that represents steady state for the tiny internet. Solve for the eigenvector $\vec{r}$ where $M\vec{r} = 1\vec{r}$ (Eigenvalue $\lambda = 1$). The "Importance Score" of all webpages is stored in a vector $\vec{r}$.
- Normalize the vector so the total traffic = 1 (or 100%) and find the steady state probability of people visiting each site.